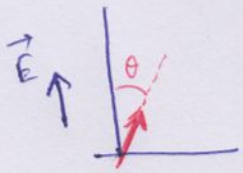
Solution

1 Sol^n Suppose the electric field is uniform. Then the net force on the dipole $\vec{p}$ is zero. So it can be slid in from infinity to the desired location without requiring any work. We can now examine the torque experienced by $\vec{p}$ because of $\vec{E}$. It will try to align $\vec{p}$ in its direction. So we must perform work to counterbalance this. This work done gets stored as the potential energy.

For small angular displacement $d\theta$,

$$dU = \underbrace{N}_{\text{Torque}}\, d\theta = pE \sin\theta \, d\theta$$



$(\vec{N} = \vec{p} \times \vec{E})$

$\therefore$ For initial θ_i to final θ_f,

$$\int_{U_i}^{U_f} dU = \int_{\theta_i}^{\theta_f} pE \sin\theta \, d\theta \Rightarrow U_f - U_i = pE \left| -\cos\theta \right|_{\theta_i}^{\theta_f}$$

$$\Rightarrow \quad U_f - U_i = -pE(\cos\theta_f - \cos\theta_i)$$

Let $U_i = 0$ at $\theta_i = \pi/2$ (U is arbitrary upto a constant)

& $U_f = U$ at $\theta_f = \theta$

$$\Rightarrow \quad U = -pE\left(\cos\theta - \cos\frac{\pi}{2}\right) = -pE\cos\theta$$

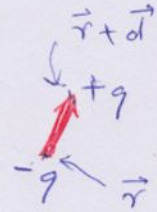
$$\Rightarrow \quad U = -\vec{p}\cdot\vec{E}$$

(Note that U is minimum for $\theta = 0$, the equilibrium (stable) position)

Holds for nonuniform $\vec{E}$ as well.

Alternative :-

Energy stored = Work done to bring $-q$ to $\vec{r}$ and $+q$ to $\vec{r}+\vec{d}$



$$\Rightarrow U = -q\,V(\vec{r}) + q\,V(\vec{r}+\vec{d})$$

$\because |\vec{d}|$ is very small (For ideal dipole $\vec{d}\to 0$)

$$V(\vec{r}+\vec{d}) = V(\vec{r}) + \vec{d}\cdot\vec{\nabla}V(\vec{r})$$

(Taylor expansion)

$$\therefore\ U = -q\cancel{V(\vec{r})} + q\cancel{V(\vec{r})} + q\vec{d}\cdot\vec{\nabla}V(\vec{r})$$

or $\quad U = \vec{p}\cdot\vec{\nabla}V(\vec{r})$

$$\Rightarrow\ U = -\vec{p}\cdot\vec{E} \qquad (\because\ \vec{E} = -\vec{\nabla}V(\vec{r}))$$

(We didn't assume uniform $\vec{E}$ in this derivation).

OR

Consider the vector identity (Griffiths, Product rule for gradient)

$$\vec{\nabla}(\vec{A}\cdot\vec{B}) = \vec{A}\times(\vec{\nabla}\times\vec{B}) + \vec{B}\times(\vec{\nabla}\times\vec{A}) + (\vec{A}\cdot\vec{\nabla})\vec{B} + (\vec{B}\cdot\vec{\nabla})\vec{A}$$

Consider $\vec{A} = \vec{p}$ & $\vec{B} = \vec{E}$

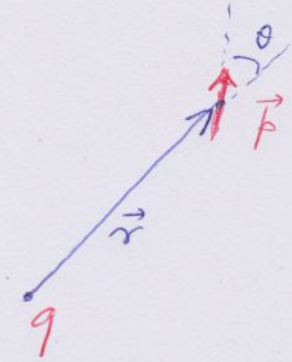
$$\Rightarrow\ \vec{\nabla}(\vec{p}\cdot\vec{E}) = \vec{p}\times(\vec{\nabla}\times\vec{E}) + \vec{E}\times(\vec{\nabla}\times\vec{p}) + (\vec{p}\cdot\vec{\nabla})\vec{E} + (\vec{E}\cdot\vec{\nabla})\vec{p}$$

Electrostatics $\Rightarrow \vec{\nabla}\times\vec{E} = 0$, Constant $\vec{p} \Rightarrow \vec{\nabla}\times\vec{p} = 0$ & $(\vec{E}\cdot\vec{\nabla})\vec{p} = 0$

$$\Rightarrow\ \vec{\nabla}(\vec{p}\cdot\vec{E}) = \underbrace{(\vec{p}\cdot\vec{\nabla})\vec{E}}_{\text{Force on } \vec{p} \text{ in nonuniform field}}$$

$$\Rightarrow\ \vec{F} = \vec{\nabla}(\vec{p}\cdot\vec{E}) \equiv -\vec{\nabla}U \quad\Rightarrow\quad \boxed{U = -\vec{p}\cdot\vec{E}}$$

Potential energy



2 Sol^n (a) To find the force on $\vec{p}$ we must find the electric field in its vicinity. Clearly this field is due to q :-

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q\vec{r}}{r^3} = \frac{q}{4\pi\epsilon_0}\frac{x\hat{i}+y\hat{j}+z\hat{k}}{(x^2+y^2+z^2)^{3/2}}$$

$$\equiv E_x\hat{i}+E_y\hat{j}+E_z\hat{k} \quad \text{Nonuniform.} \qquad (1)$$

We know that force on a dipole due to a nonuniform field is,

$$\vec{F} = (\vec{p}\cdot\vec{\nabla})\vec{E}$$

$$= \left(p_x\frac{\partial}{\partial x}+p_y\frac{\partial}{\partial y}+p_z\frac{\partial}{\partial z}\right)(E_x\hat{i}+E_y\hat{j}+E_z\hat{k})$$

$$= \hat{i}\left(p_x\frac{\partial}{\partial x}+p_y\frac{\partial}{\partial y}+p_z\frac{\partial}{\partial z}\right)E_x$$
$$+\hat{j}\left(p_x\frac{\partial}{\partial x}+p_y\frac{\partial}{\partial y}+p_z\frac{\partial}{\partial z}\right)E_y$$
$$+\hat{k}\left(p_x\frac{\partial}{\partial x}+p_y\frac{\partial}{\partial y}+p_z\frac{\partial}{\partial z}\right)E_z$$

$$\equiv \hat{i}F_x+\hat{j}F_y+\hat{k}F_z. \qquad (2)$$

We have,

$$F_x = \left(p_x\frac{\partial}{\partial x}+p_y\frac{\partial}{\partial y}+p_z\frac{\partial}{\partial z}\right)\frac{q}{4\pi\epsilon_0}\frac{x}{(x^2+y^2+z^2)^{3/2}}$$

[Using (1)]

or, $$F_x = \frac{q}{4\pi\epsilon_0}\left[p_x\left\{\frac{1}{(x^2+y^2+z^2)^{3/2}} - \frac{3}{2}x\cdot\frac{2x}{(x^2+y^2+z^2)^{5/2}}\right\}\right.$$

$$\left. + p_y\left\{-\frac{3}{2}x\ \frac{2y}{(x^2+y^2+z^2)^{5/2}}\right\} + p_z\left\{-\frac{3}{2}x\ \frac{2z}{(x^2+y^2+z^2)^{5/2}}\right\}\right]$$

or, $$F_x = \frac{q}{4\pi\epsilon_0}\left[\frac{p_x}{(x^2+y^2+z^2)^{3/2}} - \frac{3x}{(x^2+y^2+z^2)^{5/2}}(p_x x + p_y y + p_z z)\right]$$

$$= \frac{q}{4\pi\epsilon_0}\left[\frac{p_x}{r^3} - \frac{3x}{r^5}(\vec{p}\cdot\vec{r})\right]$$

$$= \frac{q}{4\pi\epsilon_0}\left[\frac{p_x}{r^3} - \frac{3x}{r^4}(\vec{p}\cdot\hat{r})\right] \quad \text{—(4)}$$

|| ly $$F_y = \frac{q}{4\pi\epsilon_0}\left[\frac{p_y}{r^3} - \frac{3y}{r^4}(\vec{p}\cdot\hat{r})\right] \quad \text{—(5)}$$

& $$F_z = \frac{q}{4\pi\epsilon_0}\left[\frac{p_z}{r^3} - \frac{3z}{r^4}(\vec{p}\cdot\hat{r})\right] \quad \text{—(6)}$$

$$\therefore \vec{F} = \frac{q}{4\pi\epsilon_0}\left[\frac{p_x\hat{i} + p_y\hat{j} + p_z\hat{k}}{r^3} - \frac{3(x\hat{i} + y\hat{j} + z\hat{k})(\vec{p}\cdot\hat{r})}{r^4}\right]$$

$$= \frac{q}{4\pi\epsilon_0}\left[\frac{\vec{p}}{r^3} - \frac{3\vec{r}(\vec{p}\cdot\hat{r})}{r^4}\right]$$

$$= \frac{q}{4\pi\epsilon_0 r^3}\left[\vec{p} - 3(\vec{p}\cdot\hat{r})\hat{r}\right] \quad \text{—(7)}$$

(b) To find the force on q we should know the external field at its location. Clearly this field is due to the dipole, which is,

$$\vec{E}' = \frac{1}{4\pi\epsilon_0 r^3}\left[3(\vec{p}\cdot(-\hat{r}))(-\hat{r}) - \vec{p}\right]$$

or, $$\vec{E}' = \frac{1}{4\pi\epsilon_0 r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$$

(Remains same)

The electric field due to a dipole at a position measured with respect to it is $\vec{E} = \frac{1}{4\pi\epsilon_0 r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$. In the present case position is measured with respect to q. So $\hat{r} \to -\hat{r}$.

$\therefore$ Force on q,

$$\vec{F}' = q\vec{E}'$$

$$= \frac{q}{4\pi\epsilon_0 r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right] \quad (8)$$

Comparing (7) and (8), we see that $\vec{F} = -\vec{F}'$, as expected from Newton's third law.

3 Soln

The surface bound charge density is given by,

$$\sigma_b = \vec{P} \cdot \hat{n} \quad \text{at the surface.}$$

Here $\hat{n} = \hat{r}$, the unit vector in the radial direction. Also at the surface of the sphere $r = R$.

$$\therefore \sigma_b = k r \vec{r} \cdot \hat{r} \Big|_{r=R}$$

$$= k r^2 \hat{r} \cdot \hat{r} \Big|_{r=R}$$

$$= kR^2$$

Volume bound charge density is given by,

$$\rho_b = -\vec{\nabla} \cdot \vec{P}$$

We know that $\vec{\nabla} \cdot \vec{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r)$ when $\vec{v}$ has only r dependence i.e. $\vec{v} = v_r \hat{r}$.

This is the case with $\vec{P}$.

$$\vec{P} = k r \vec{r} = k r^2 \hat{r}$$

$$\therefore P_r = k r^2$$

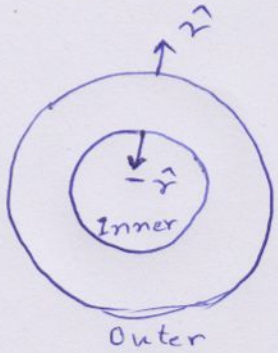
$$\therefore \rho_b = -\vec{\nabla} \cdot \vec{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \, k r^2)$$

or, $\rho_b = -\frac{k}{r^2} (4 r^3) = -4kr$

Varies in the volume.

4 Sol^n (a) Bound charges :-

Surface :- There are two surfaces



• Outer : $r = b$ & $\hat{n} = \hat{r}$

$\therefore \sigma_B = \vec{P} \cdot \hat{n}$ at $r = b$

Using capital B to avoid confusion with radius b

$$\Rightarrow \sigma_B = \frac{k}{r} \hat{r} \cdot \hat{r} \Big|_{r=b}$$

or, $$\sigma_B = \frac{k}{b} \quad \text{—(1)}$$

• Inner : $r = a$ & $\hat{n} = -\hat{r}$

$\therefore \sigma_B = \vec{P} \cdot \hat{n}$ at $r = a$

$$\Rightarrow \sigma_B = \frac{k}{r} \hat{r} \cdot (-\hat{r}) \Big|_{r=a}$$

or, $$\sigma_B = -\frac{k}{a} \quad \text{—(2)}$$

Volume :- $\rho_B = -\vec{\nabla} \cdot \vec{P}$

$$\Rightarrow \rho_B = -\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{k}{r}\right)$$ [Only r dependence]

or, $$\rho_B = -\frac{k}{r^2} \quad \text{—(3)}$$

Applying Gauss's law (in terms of $\vec{E}$) in the three regions :-

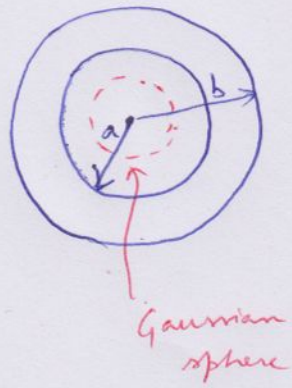
- $r < a$ $\qquad \epsilon_0 \oint \vec{E} \cdot d\vec{S} = Q_{total,\,enc} = 0$

No charge in this region

$\Rightarrow \epsilon_0 E \oint dS = 0$

$\Rightarrow E = 0$

$\Rightarrow \vec{E} = 0$ for $0 \leq r < a$



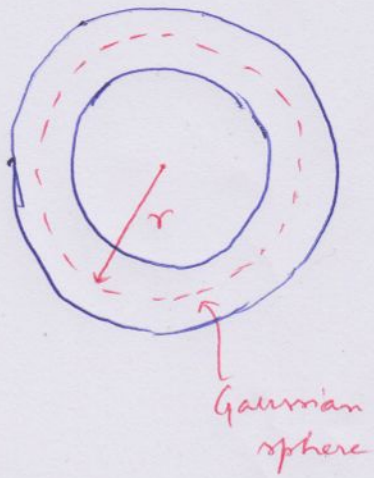
- $a < r < b$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = Q_{free,\,enc} + Q_{B,\,enc}$$

$$= Q_{B,\,enc} \qquad (\because Q_{free} = 0, \text{ given})$$

$$\Rightarrow \epsilon_0 E (4\pi r^2)$$

$$= \int_{\text{(Inner Surface)}} \sigma_B \, dS' + \int \rho_B \, d\tau'$$



$$\Rightarrow 4\pi\epsilon_0 r^2 E = -\int_{\text{(Inner Surface)}} \frac{k}{a} \, dS'$$

$$+ \int_0^a (0) \, d\tau' + \int_a^r \left(-\frac{k}{r'^2}\right)(4\pi r'^2) \, dr'$$

$\because \rho_B = 0$ for $r < a$ $\qquad$ (Using (2) & (3))

or, $$4\pi\epsilon_0 r^2 E = -\frac{k}{a}\int_{r=a} dS' - 4\pi k \int_a^r dr'$$

$$\Rightarrow 4\pi\epsilon_0 r^2 E = -\frac{k}{a} 4\pi a^2 - 4\pi k (r-a)$$

$$= -\cancel{4\pi} ka - 4\pi k r + \cancel{4\pi} ka$$

$$\Rightarrow \vec{E} = -\left(\frac{k}{\epsilon_0 r}\right)\hat{r}, \quad \text{for } a < r < b$$

- $r > b$

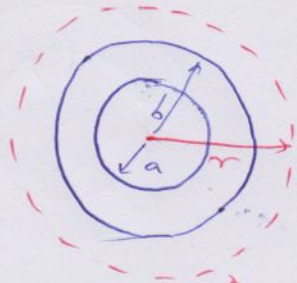
$$\epsilon_0 \oint \vec{E}\cdot d\vec{S} = Q_{B,enc} \quad (\because Q_{free} = 0)$$

$$\Rightarrow \epsilon_0 E (4\pi r^2) = \int_{\text{(inner surface)}} \sigma_B\, dS' + \int \rho_B\, d\tau' + \int_{\text{(outer surface)}} \sigma_B\, dS'$$

$$\Rightarrow 4\pi\epsilon_0 r^2 E = -\frac{k}{a}(4\pi a^2)$$

$$+ \int_a^b \left(-\frac{k}{r'^2}\right)(4\pi r'^2)\, dr'$$

$$+ \frac{k}{b}(4\pi b^2) \qquad \text{(Using (1), (2), (3))}$$



Gaussian sphere

$$= -4\pi ka - 4\pi k(b-a) + 4\pi kb$$

$$= 0$$

$$\Rightarrow \vec{E} = 0 \quad \text{for } r > b$$

(b) We'll now use Gauss's law expressed in terms of $\vec{D}$:

$$\oint \vec{D} \cdot d\vec{S} = Q_{free,enc} = 0 \qquad (\because Q_{free} = 0, \text{ given})$$

$\Rightarrow$ $D\,(4\pi r^2) = 0$ for any choice of Gaussian sphere

$\Rightarrow$ $D = 0$.

$\Rightarrow$ $\vec{D} = 0$ everywhere.

$\Rightarrow$ $\epsilon_0 \vec{E} + \vec{P} = 0 \Rightarrow \vec{E} = -\frac{1}{\epsilon_0}\vec{P}$.

- For $0 \le r < a$, $\vec{P} = 0 \Rightarrow \vec{E} = 0$

(Only the thick shell is polarized)

- For $a < r < b$, $\vec{P} = \frac{k}{r}\hat{r} \Rightarrow \vec{E} = -\frac{k}{\epsilon_0 r}\hat{r}$

- For $r > b$, $\vec{P} = 0 \Rightarrow \vec{E} = 0$.

It was much easier to calculate the electric field using this approach!

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